Hence $x = \frac{\Delta x}{\Delta} = \frac{-44}{-11} = 4$, $y = \frac{\Delta y}{\Delta} = \frac{11}{-11} = -1$

$\therefore x = 4$ and $y = -1$

Answer checking:

Substituting in equation the values of x and y,

4 + 3(-1) = 1,

3(4) – 2(-1)= 14

Example: 12.9

Find the solution of the system of equations.

$$5x_1 + 3x_2 = 30$$

$$6x_1 - 2x_2 = 8$$

Solution:

The coefficient and the constant terms are given below for the equations

$$\Delta = \begin{vmatrix} 5 & 3 \\ 6 & -2 \end{vmatrix} = -10 - 18 = -28$$

$$\Delta x_1 = \begin{vmatrix} 30 & 3 \\ 8 & -2 \end{vmatrix} = -60 - 24 = -84$$

$$\Delta x_2 = \begin{vmatrix} 5 & 30 \\ 6 & 8 \end{vmatrix} = +40 - 180 = -140$$

$$\therefore x_1 = \frac{\Delta x_1}{\Delta} = \frac{-84}{-28} = 3$$

$$x_2 = \frac{\Delta x_2}{\Delta} = \frac{-140}{-28} = 5$$

$\therefore x_1 = 3, \quad x_2 = 5$

Example: 12.10

Find the solution of the system of equation

$$7x_1 - x_2 - x_3 = 0$$

$$10x_1 - 2x_2 + x_3 = 8$$

$$6x_1 + 3x_2 - 2x_3 = 7$$

Solution:

The matrix form of the given equation is written as

$$\begin{bmatrix} 7 & -1 & -1 \\ 10 & -2 & 1 \\ 6 & 3 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 8 \\ 7 \end{bmatrix}$$

$$\Delta = \begin{vmatrix} 7 & -1 & -1 \\ 10 & -2 & +1 \\ 6 & 3 & -2 \end{vmatrix}$$

$$= 7(4-3) - (-1)(-20-6) + (-1)(30+12)$$

$$= 7(1) + 1(-26) - 1(42)$$

$$= 7 - 26 - 42 = -61$$

$$\Delta x_1 = = \begin{vmatrix} 0 & -1 & -1 \\ 8 & -2 & 1 \\ 7 & 3 & -2 \end{vmatrix}$$

$$= 0(4-3) - (-1)(-16-7) + (-1)(24+14)$$

$$= 0 + 1(-23) - 1(38)$$

$$= -23 - 38 = -61$$

$$\Delta x_2 = = \begin{vmatrix} 7 & 0 & -1 \\ 10 & 8 & 1 \\ 6 & 7 & -2 \end{vmatrix}$$

$$= 7(-16-7) - 0(-20-6) + (-1)(70-48)$$

$$= 7(-23) + 0 - 1(22)$$

$$= -161 - 22 = -183$$

$$\Delta x_3 = = \begin{vmatrix} 7 & -1 & 0 \\ 10 & -2 & 8 \\ 6 & 3 & 7 \end{vmatrix}$$

$$= 7(-14-24) - (-1)(70-48) + 0(30+12)$$

$$= 7(-38) + 1(22) + 0$$

$$= -266 + 22 = -244$$

$$x_1 = \frac{\Delta x_1}{\Delta} = \frac{-61}{-61} = 1$$

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